BCS-085	Big Data Analysis	L T P 3 0 0	3 Credits

Pre-requisites: Students should know about the data base.
Students should know about the data mining.

Course Objectives:

CO1	Provide an overview of Apache Hadoop
CO2	Provide HDFS Concepts and Interfacing with HDFS
CO3	Understand Map Reduce Jobs
CO4	Apply analytics on Structured, Unstructured Data.
CO5	Understand the analysis of Big data

Detailed Syllabus

MODULE-I

Introduction – distributed file system – Big Data and its importance, Four Vs, Drivers for Big data, Big data analytics, Big data applications. Algorithms using map reduce, Matrix- Vector Multiplication by Map Reduce INTRODUCTION HADOOP Big Data – Apache Hadoop & Hadoop EcoSystem – Moving Data in and out of Hadoop.

MODULE 2

HADOOP ARCHITECTURE Hadoop Architecture, Hadoop Storage: HDFS, Common Hadoop Shell commands, Anatomy of File Write and Read., NameNode, Secondary NameNode, and DataNode, Hadoop MapReduce paradigm, Map and Reduce tasks, Job, Task trackers - Cluster Setup – SSH & Hadoop Configuration – HDFS Administering –Monitoring & Maintenance

MODULE 3

HADOOP ECOSYSTEM AND YARN Hadoop ecosystem components - Schedulers – Fair and Capacity, Hadoop 2.0 New Features Name Node High Availability, HDFS Federation, MRv2, YARN, Running MRv1 in YARN, Hive Architecture and Installation, Comparison with Traditional Database, HiveQL - Querying Data - Sorting And Aggregating.

Course Outcomes: After the completion of the course the student will be able to:

CO1	Analyze Infosphere BigInsights Big Data Recommendations.
CO2	Manage Job Execution in Hadoop Environment
CO3	Develop Big Data Solutions using Hadoop Eco System
CO4	Apply Machine Learning Techniques using R.
CO5	Analyze Big Data by using various techniques

Text Books:

1. The Big Data-Driven Business: How to Use Big Data to Win Customers, Beat Competitors, and Boost Profits Russell Glass,Sean Callahan.
2. Data Fluency: Empowering Your Organization with Effective Data Communication, Zach Gemignani,Chris Gemignani,Richard Galentino.
3. Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data,Gebundene Ausgabe,von EMC Education Services (Herausgeber)
4. Hadoop: The Definitive GuideAuthor: Tom White Publisher: Hadoop: The Definitive Guide
5. Hadoop in Action Author: Chuck Lam Publisher: Manning